

Zhenkai Zhang (Ken)

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EDUCATION

The Hong Kong Polytechnic University

Starting Sep 2026

MSc in Data Science and Analytics

Nanjing Tech University

Sep 2022 - Jun 2026

BSc in Data Science and Big Data Technology | 3.9/4.0

RESEARCH INTERESTS

Efficient AI; model routing and reuse; multimodal agents; generative AI

SELECTED PUBLICATIONS

Zhang, Z., Gao, Y., & Chen, D. (2025). Exploring and Enhancing Advanced MoE Models: From DeepSpeed-MoE to DeepSeek-V3. 2025 *IEEE 6th International Seminar on Artificial Intelligence, Networking and Information Technology (AINIT)*. First author. [DOI](#)

Zhang, Z., Ma, T., Yao, Y., et al. (2025). Predicting Olympic Medal Performance for 2028: Machine Learning Models and the Impact of Host and Coaching Effects. *Applied Sciences*, 15(14), 7793. First author. [DOI](#)

Xia, W., Gao, Y., **Zhang, Z.**, et al. (2025). Behavior Prediction of Connections in Eco-Designed Thin-Walled Steel-Ply-Bamboo Structures Based on Machine Learning for Mechanical Properties. *Sustainability*, 17(15), 6753. Third author. [DOI](#)

SELECTED PROJECTS

RouterBench-Mini: Cost-Aware Model Reuse for Multimodal Agents

Jul 2026

Independent Research Project | [GitHub](#)

- Designed and released a 4,000-example routing benchmark with a 3,200-example development set and a frozen 800-example test set spanning text, vision, and tool-use tasks.
- Compared fixed baselines, frozen Task-Aware routing, learned quality-gap routing, and post-response Reflection using five-fold OOF selection, overlap audits, ablations, paired statistics, and failure analysis.
- On the frozen test set, Task-Aware reached 72.12% accuracy (0.63 percentage points below Always Strong) while reducing average cost by 13.94% and latency by 14.01%; released manifests, per-example predictions, scripts, tests, and CI.

Finance-Agent: Multi-Agent Financial Research System

Oct 2025 - Jan 2026

Engineering Project | [GitHub](#)

- Implemented a LangGraph workflow with four specialist agents for fundamental, technical, valuation, and news analysis, followed by report synthesis, evaluation, and one bounded reflection retry.
- Standardized financial data access through 28 registered FastMCP tools across eight modules covering market data, reports, indices, macro data, analysis, and news.
- Verified 60/60 deterministic company-name and stock-code extraction cases; published the evaluation scope, a machine-readable result CSV, and an anonymized report example without presenting unverified model metrics.

TECHNICAL SKILLS

Python, PyTorch, Transformers, scikit-learn, PEFT/LoRA, LangGraph, MCP, FastAPI, Git, Linux, pytest, CI

HONORS

Meritorious Winner (Problem F), Mathematical Contest in Modeling (MCM)

Feb 2026